

ODONT 541.1

CARIOLOGY

Root Caries

Lecture #4

Sahel Farhangi, DDS
Office Hours: By appointment

Caries Detector Dye

- Ingredients:
 - 1% dye: highly visible anionic stain that adsorbs to **denatured collagen and porous organic matrix**
 - Propylene glycol: low viscosity solvent that penetrates and carries the dye deeper into dentin
 - Some brands use POLYpropylene glycol instead, which is **MORE** viscous and may not penetrate as deep.
 - Highlighted correction: changed "less" to MORE for polypropylene glycol
- LOW viscosity = watery, thin, penetrates deeper
- HIGH viscosity = thick, syrupy, doesn't penetrate as deep
 - MORE viscous = thicker



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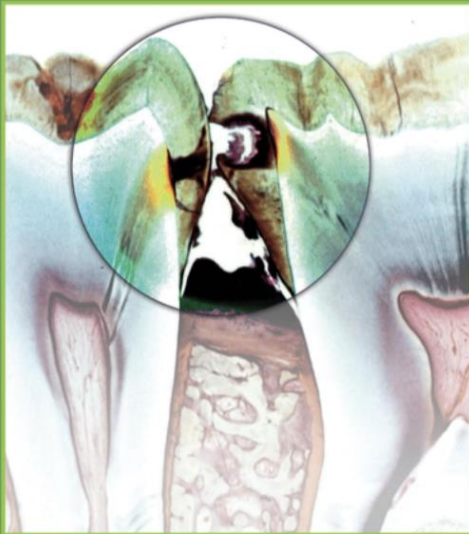
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Recommended Reading

Fourth Edition

Dental Caries

The Disease and Its Clinical Management



Edited by **Ole Fejerskov** and **Bente Nyvad**

WILEY Blackwell

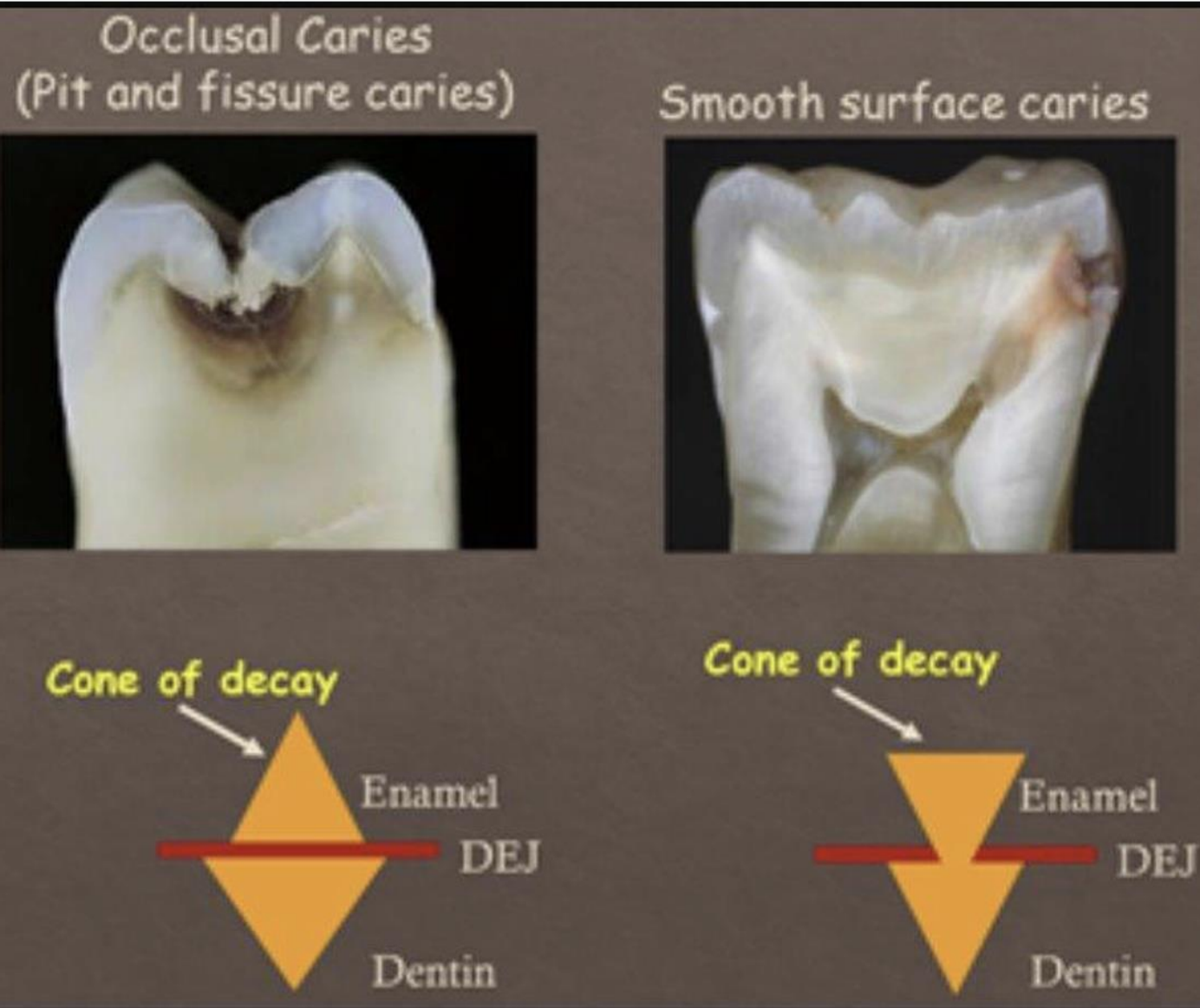
Textbook:

- **Page 227-230 : Root caries**
- **Page 233-246: Erosion**

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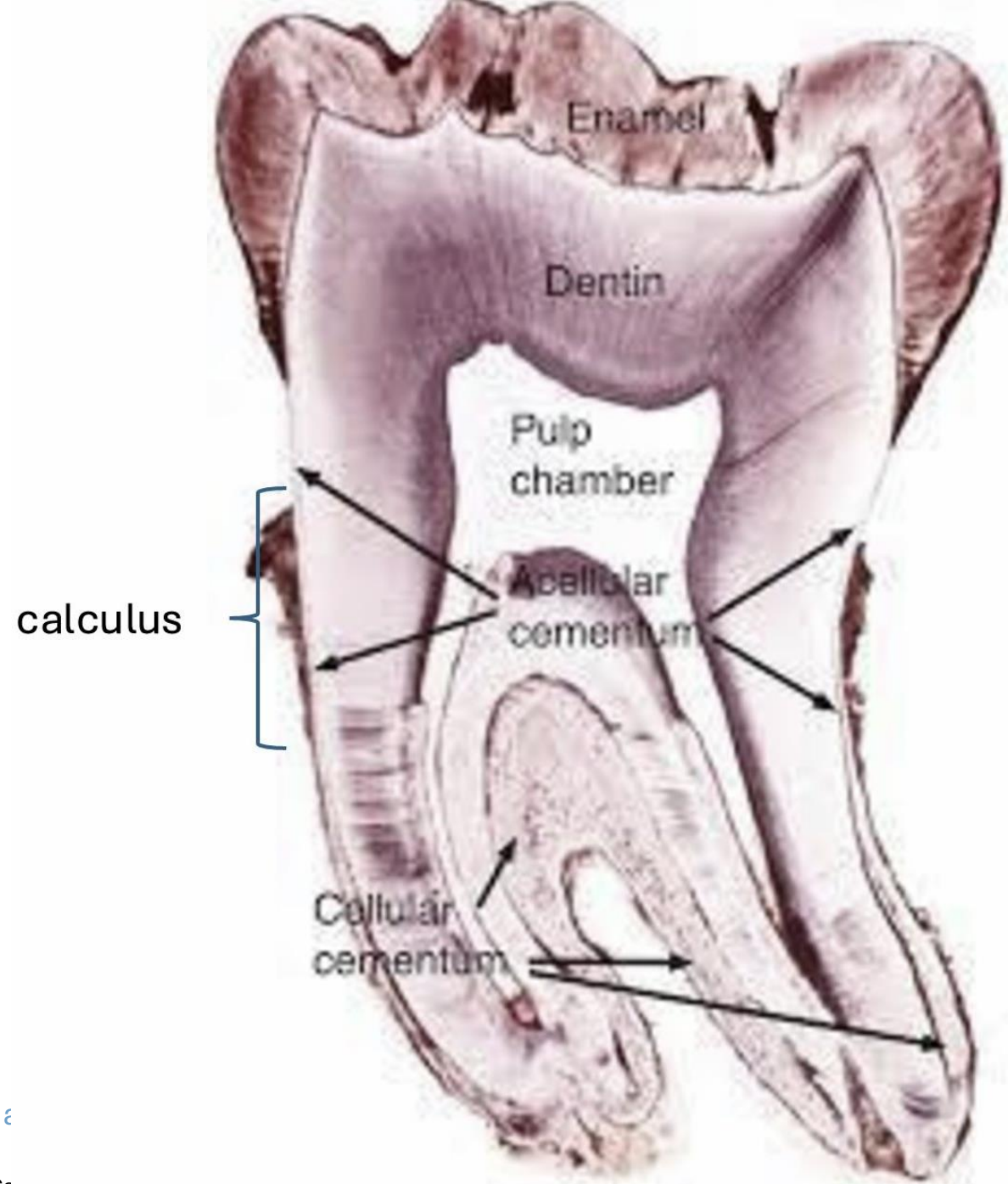
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Recap from Last Week



Cementum

- Functions:
 - Covers dentinal tubules over root surface
 - Anchors periodontal ligament fibers to the root
- Composed of
 - 45 vol% mineral
 - 55 vol% organic
 - Collagens
 - Protein polysaccharides
- Critical pH of ~~6.0-6.7~~: **6.7**



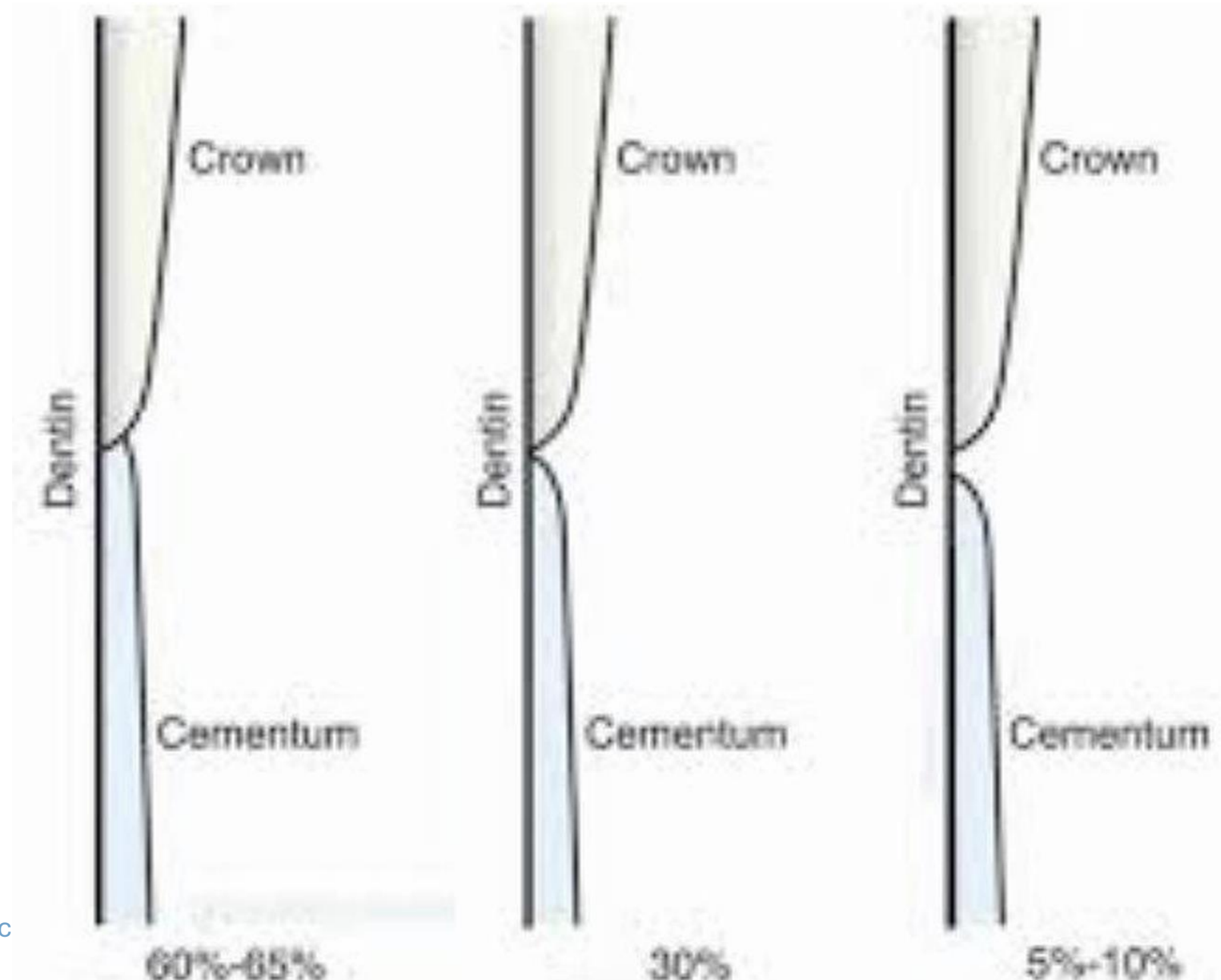
Tissue	Mineral (HA)	Organic	Critical pH
Enamel	~96%	~2%	5.5 (HA), 4.5 (FA)
Dentin	~70%	~20%	6.5
Cementum	~45-60%	~40-55%	6.7

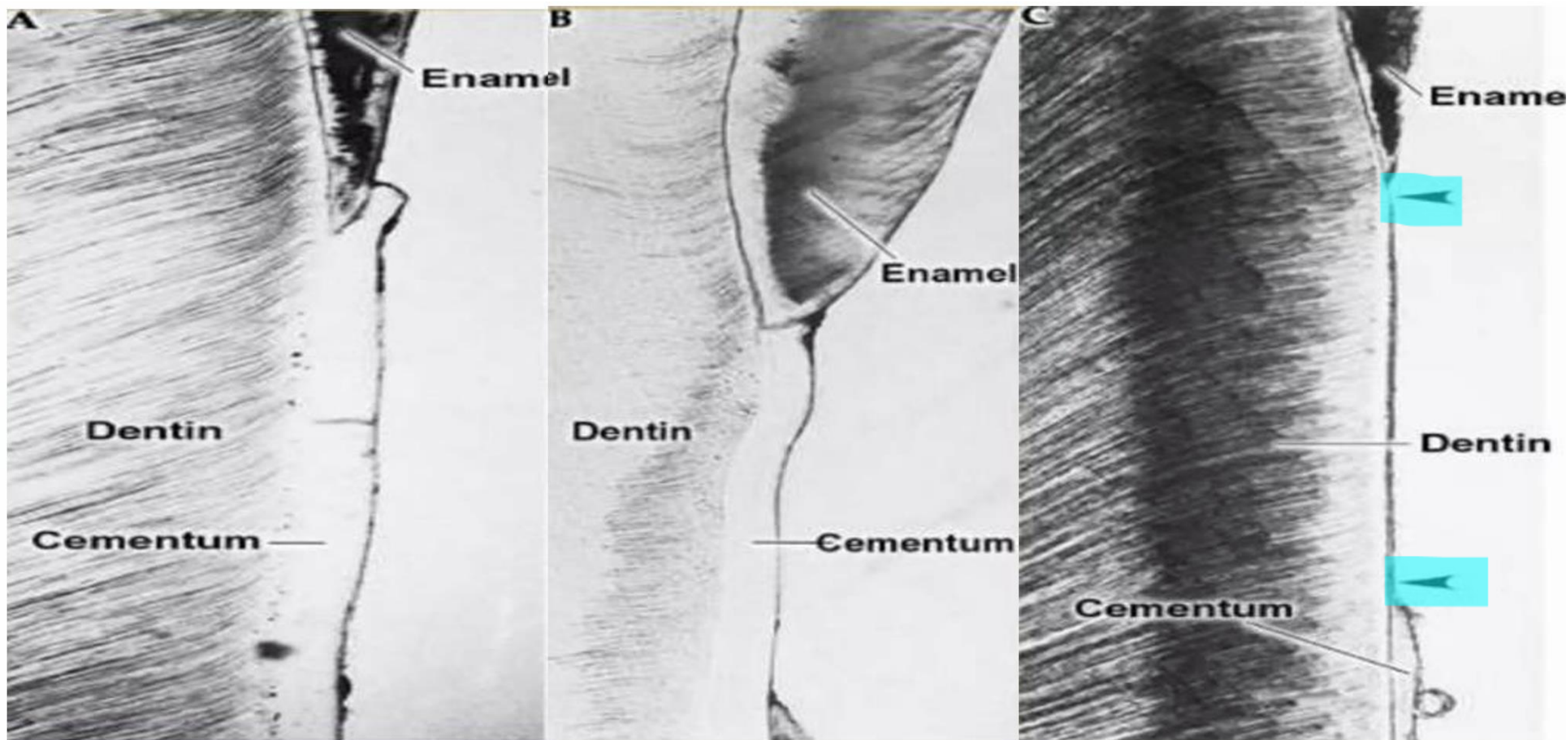
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CEJ structure

- 60-65% of the time, overlapping cementum and enamel
- 30% of the time, contacting enamel and cementum
- 5-10% of the time, "open CEJ", dentin exposed in small space between cementum and enamel.



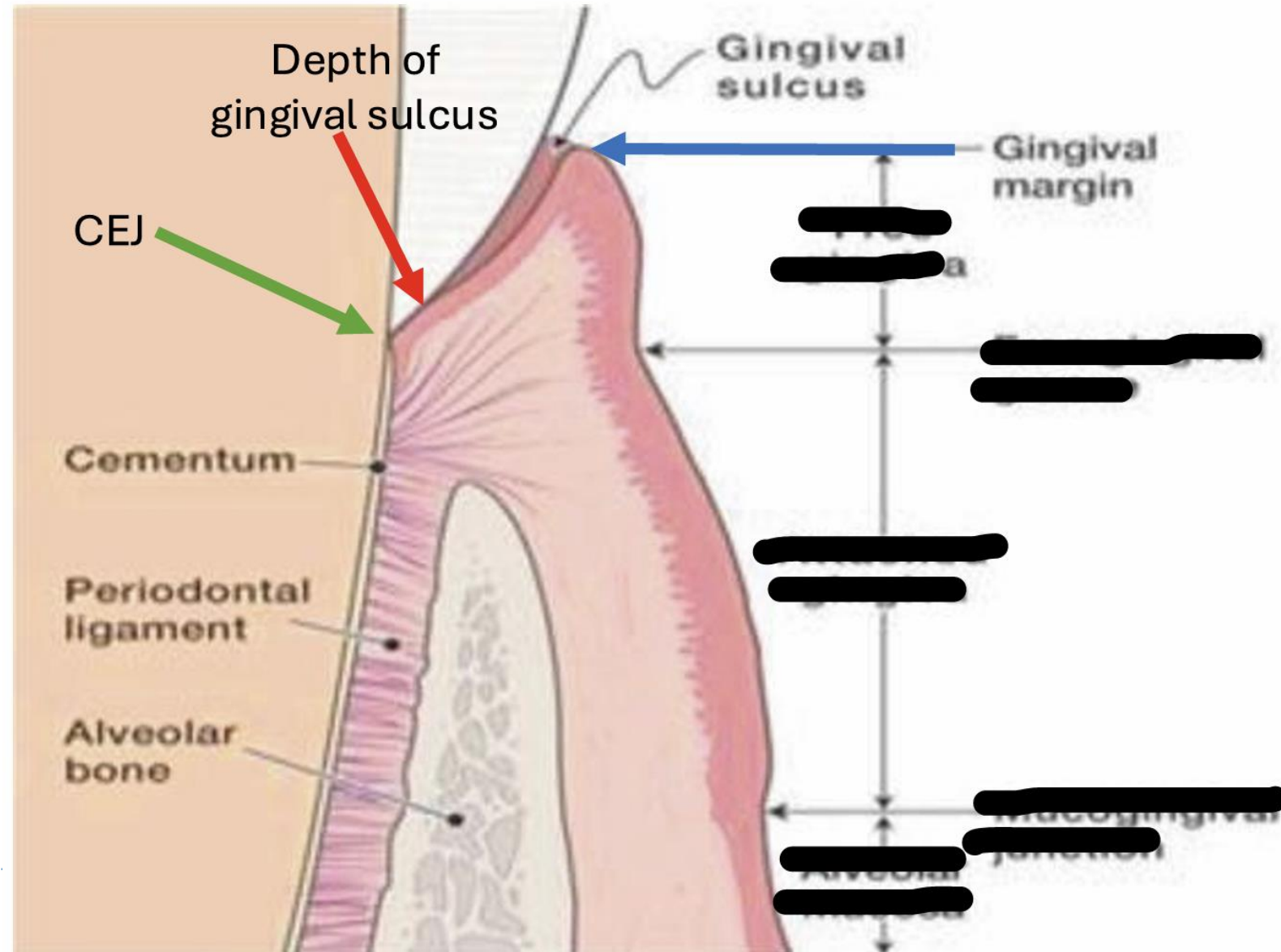


"Healthy" Periodontium

Periodontium include:

- Periodontal ligament (PDL)
- Cementum
- Alveolar bone
- Gingival tissue

Notice that in a healthy periodontium, the depth of the sulcus is coronal to the CEJ. Cementum is not exposed. Even if the CEJ were "open," it would not be exposed to stimuli as it would be periodontally protected.



Gingival Recession

Cementum/root dentin become exposed to the oral environment in cases of gingival recession.

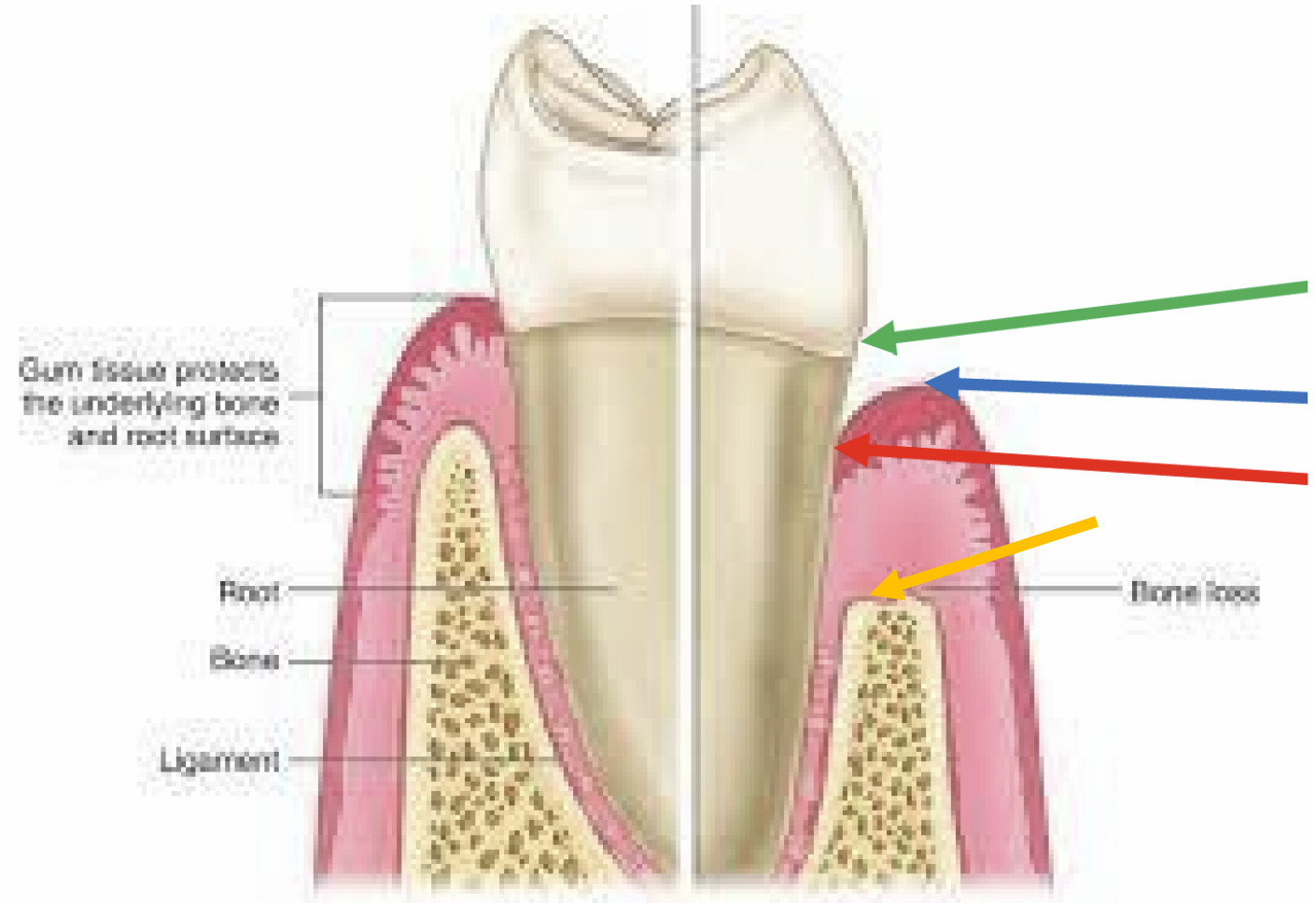
If the root is not exposed, it can not develop caries.

Green arrow: CEJ

Blue arrow: Gingival margin

Red arrow: Depth of sulcus

Orange arrow: Bone



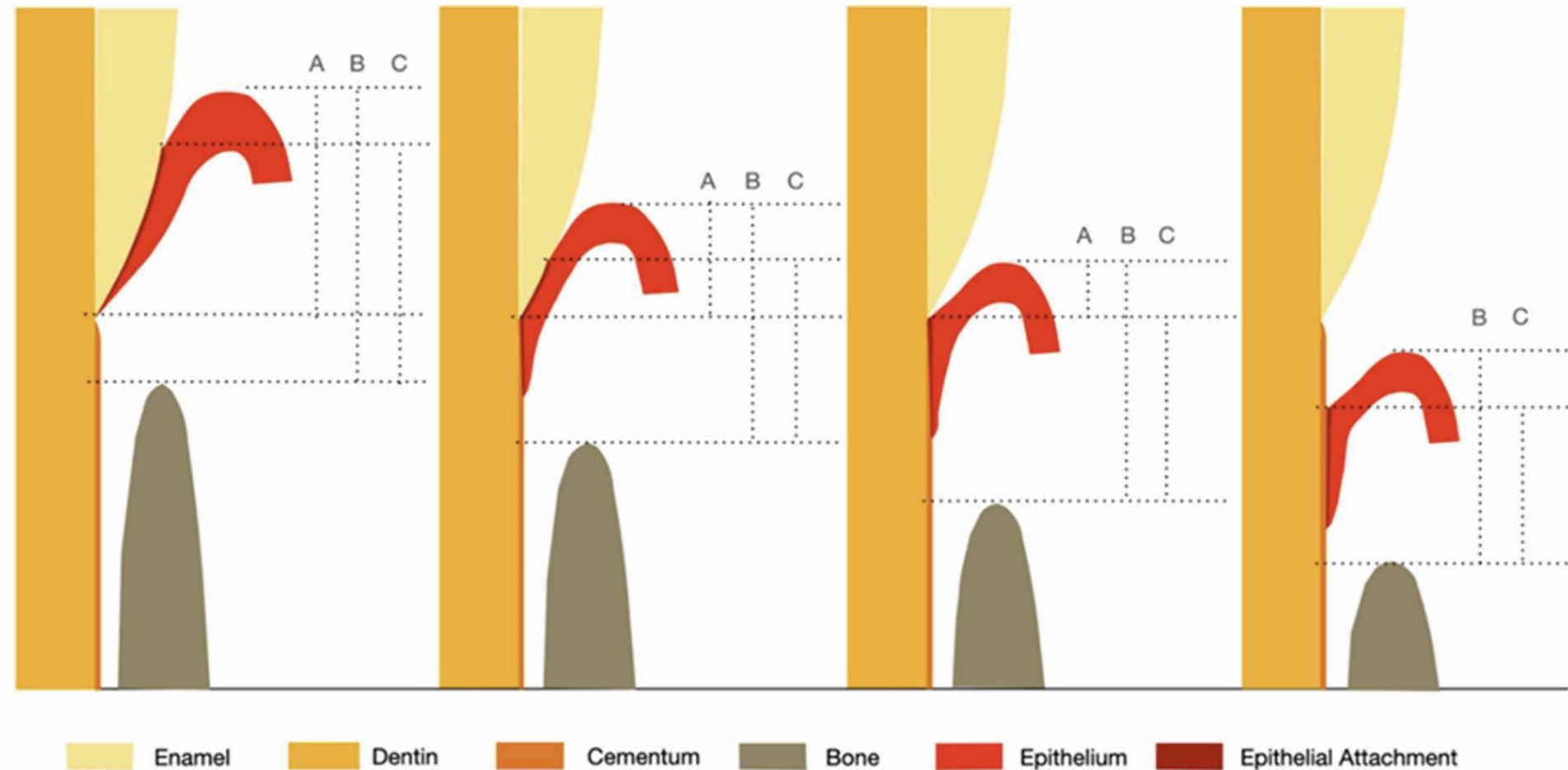
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Dental plaque often accumulates at the gingival margins.

This can lead to gingival inflammation and eventually, changes in the height of the osseous crest and gingival margin.

Demineralization also occurs where the biofilm accumulates without disruption (if pH supports demin).



Gingival Recession

Possible etiologic factors:

- Biofilm mediated inflammation
- Parafunctional forces
 - o Bruxism
- Mechanical abrasion
- Site specific trauma
 - o Piercings
 - o Injuries
- Frenal attachments
- Denture clasps
- Nearby extractions with bone loss
- Smoking
- Ectopic positioning or crowding.
- Genetic/hormonal



- By adulthood, 90% of adults have experienced caries.
- The prevalence of root caries increases with age, which correlates with an increase in:
 - Xerostomia
 - Root exposure
- Does this mean gingival recession is inevitable with age?
 - NO. But it is a cumulative effect over time.

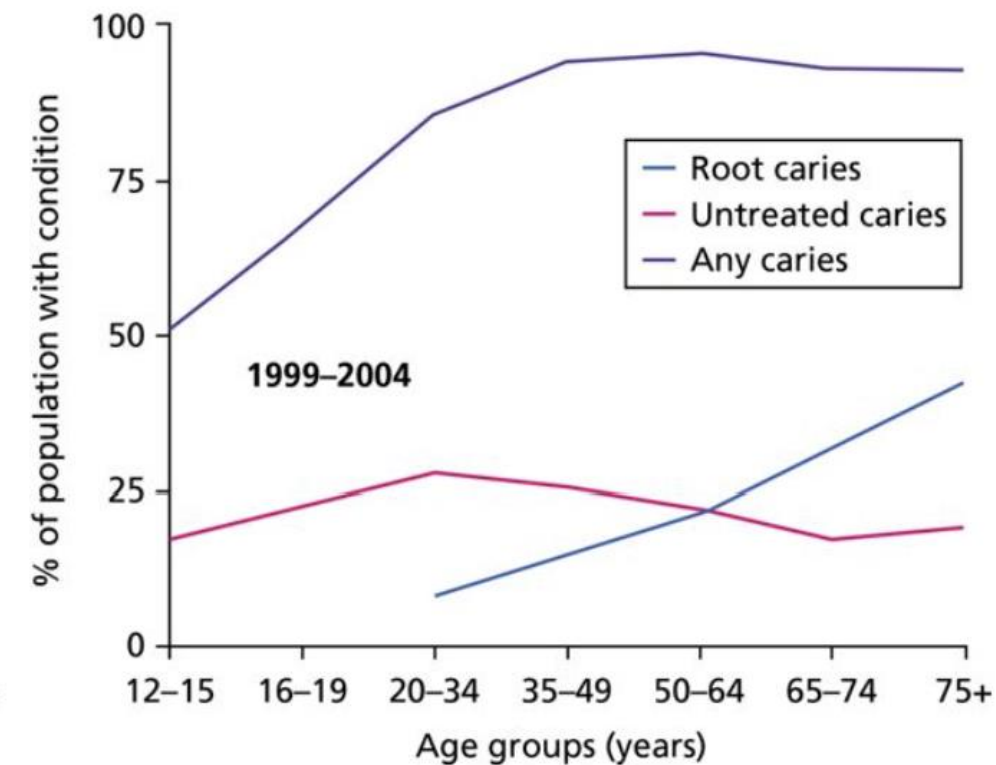
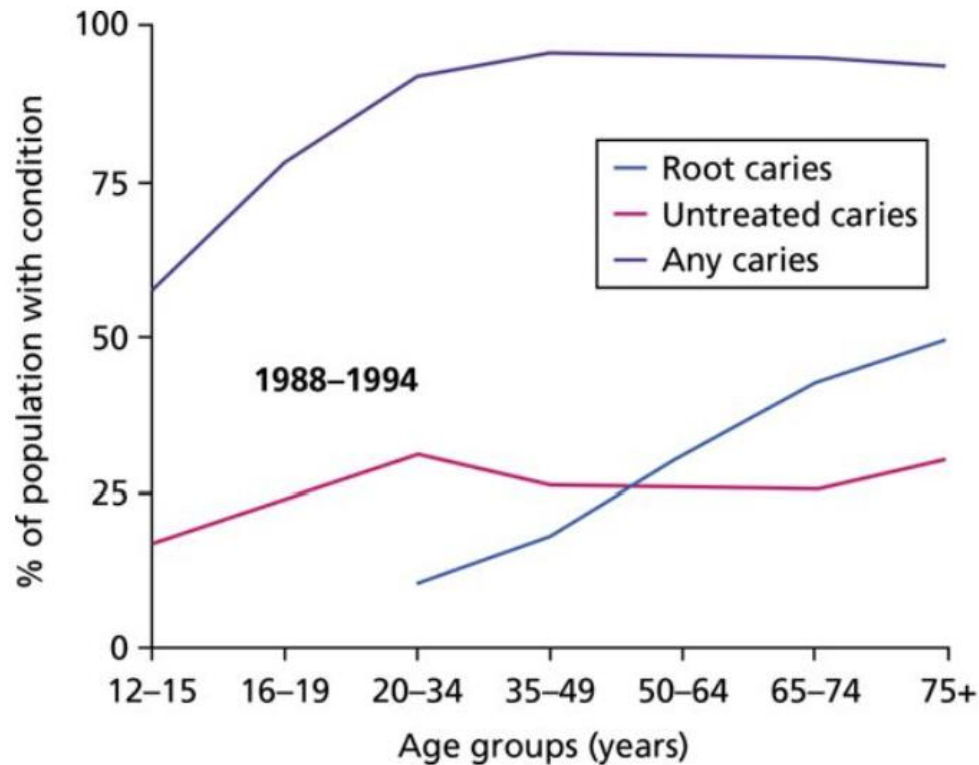


Figure 4.10 The prevalence of any caries (D₃ level), untreated caries (D₃), and root caries among US adolescents, adults, and elderly in two national surveys, one in 1988-1994 and one in 1999-2004.

Root caries often initiates at the CEJ, the height of the gingival margin, or at the margin of dental restorations.



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- The CEJ can become exposed with loss of periodontal attachment or significant passive eruption
 - o The CEJ is highly irregular and can be a significant bacterial retention site
 - o Therefore, a majority of root caries initiate at this site
- Do you think the margin of the average dental restoration is rougher or smoother than a CEJ?

The CEJ: A Biofilm and Calculus Trap

Keerthana Satheesh, BDS, DDS, MS; Simon R. MacNeill, BDS, DDS; John W. Rapley, DDS, MS; and Charles M. Cobb, DDS, MS, PhD

February 2015 RN - Expires Wednesday, February 28th, 2018

Parkell Online Learning Center

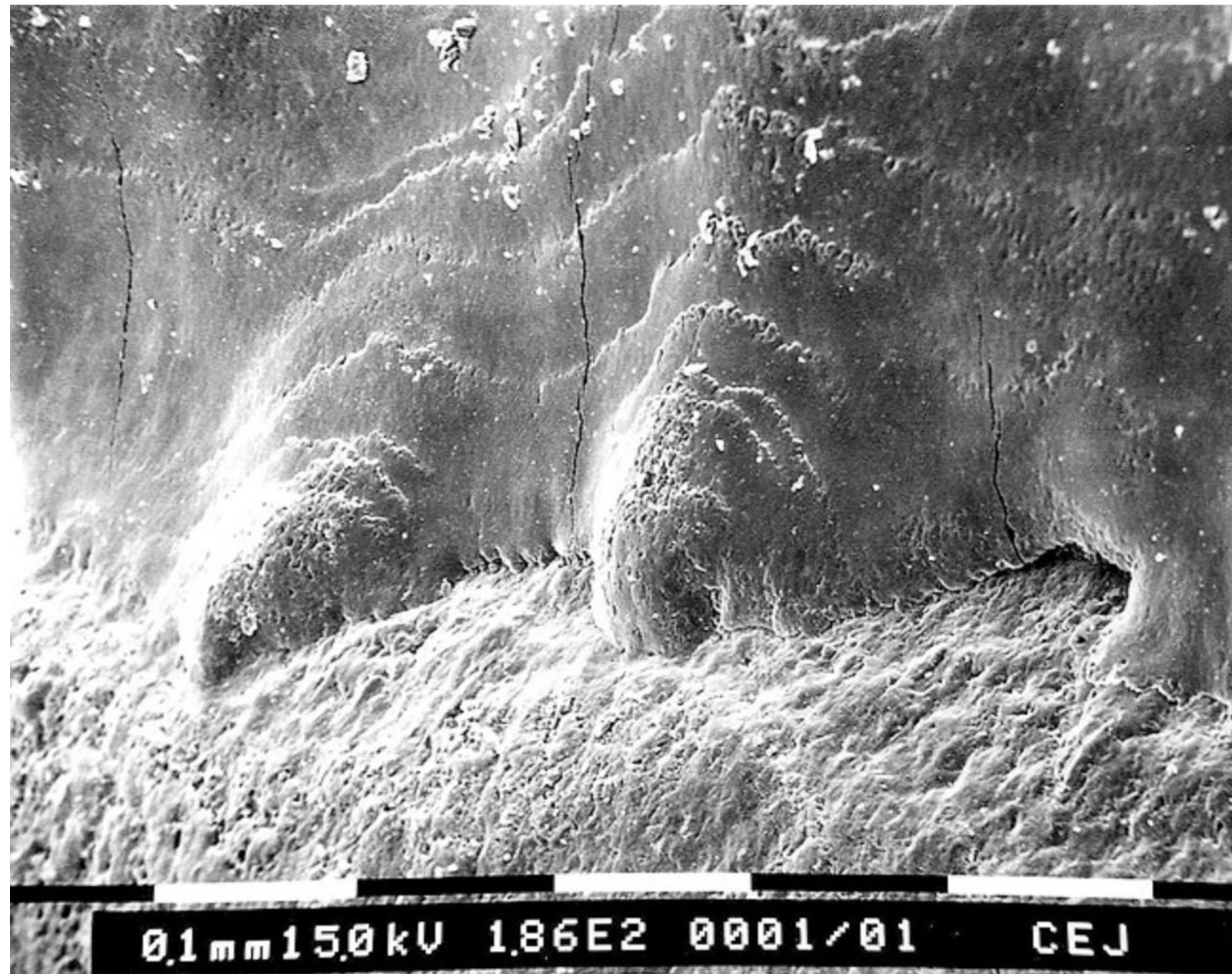


Image 4 of 10

Figure 3 Irregular contours of enamel at the CEJ, rough and prominent perikymata, and tunnel-like area where enamel overlaps the cementum (arrow) all contribute to retention of biofilm and dental calculus. Bar = 0.1 mm at an original magnification of 186x.

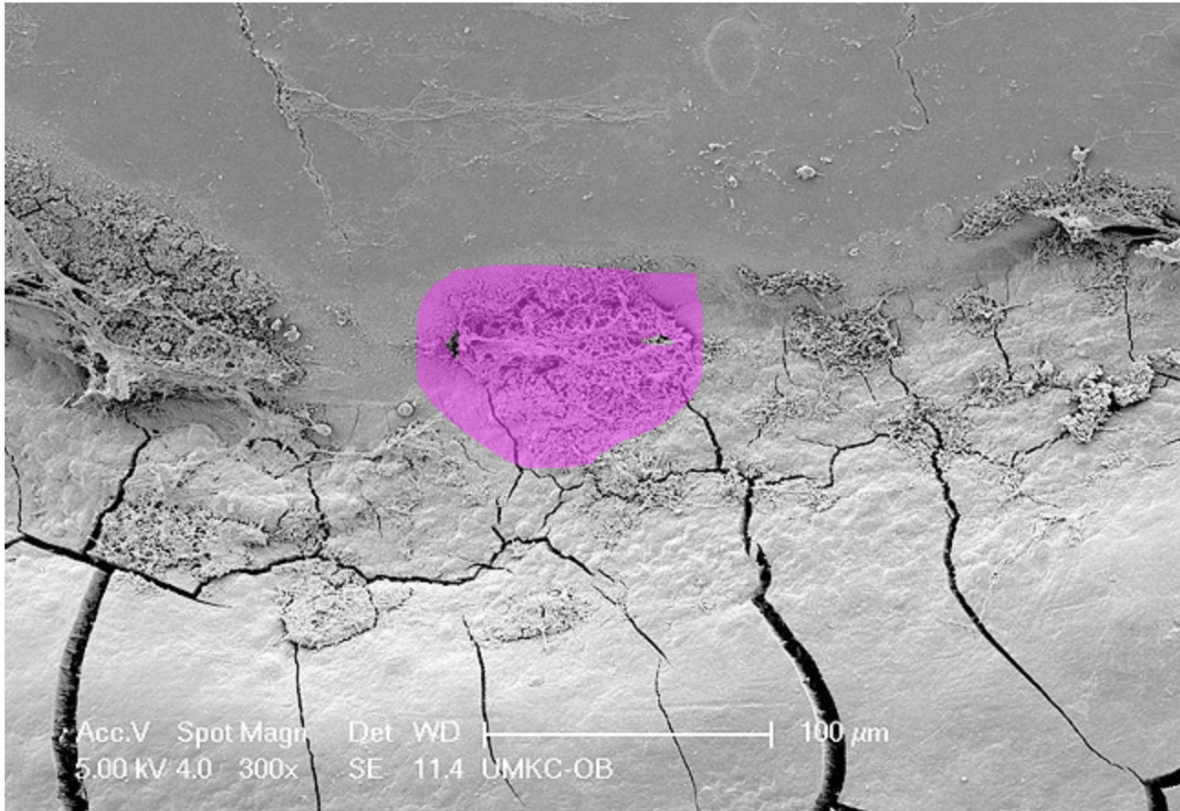


Image 6 of 10

Figure 5 Irregular contour of CEJ showing aggregations of biofilm in scalloped enamel recesses (arrows). Enamel is at top and cementum at bottom of photograph. Cracks in cementum are the result of dehydration during specimen preparation. Bar = 100 μm at an original magnification of 300x.

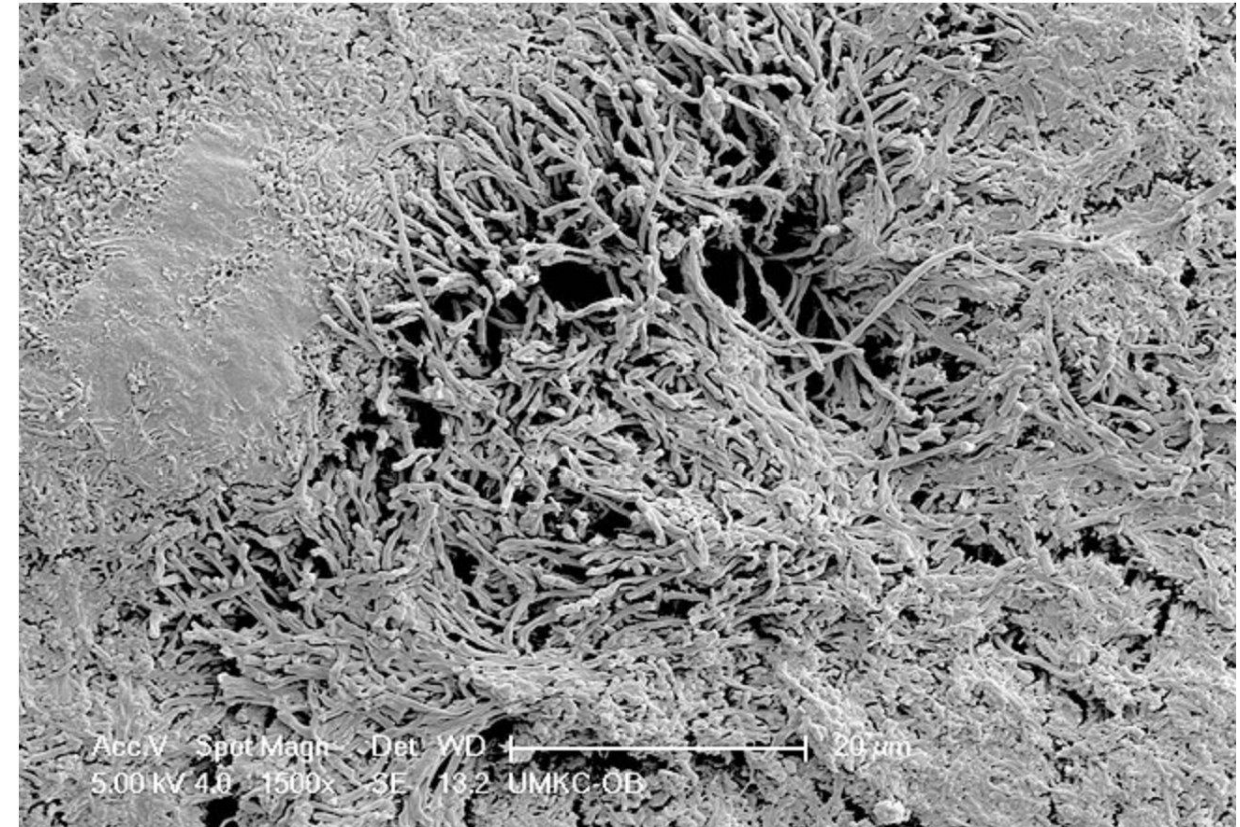


Image 7 of 10

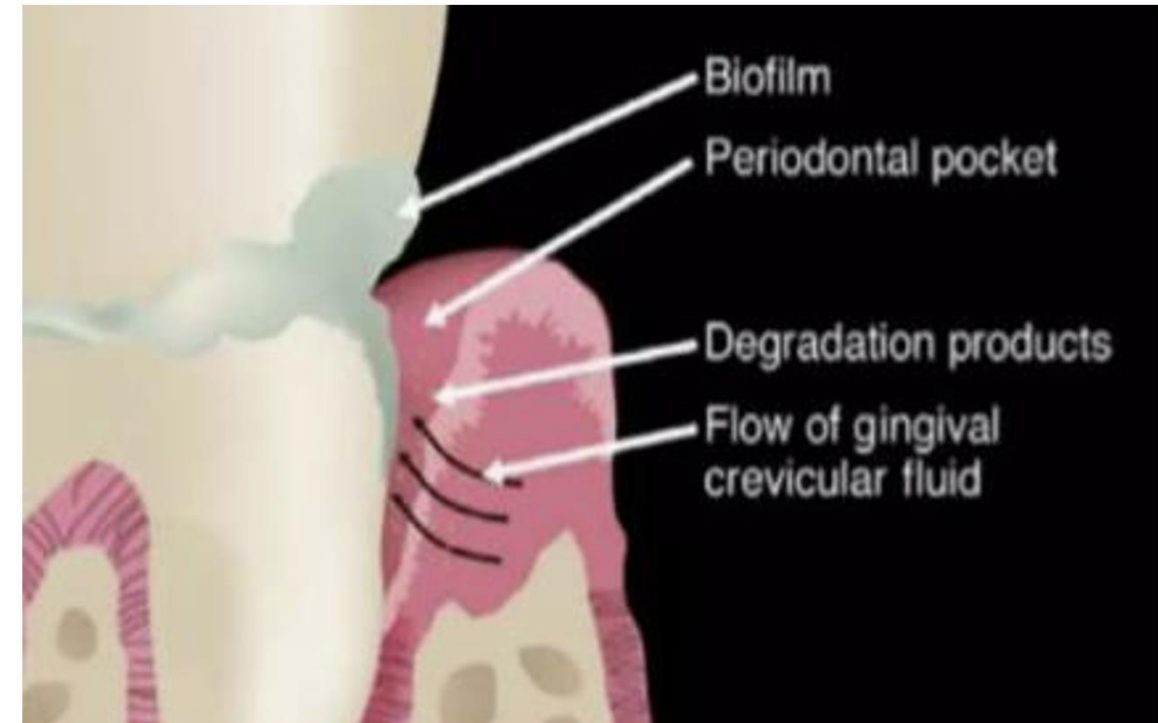
Figure 6 High-magnification view of biofilm noted in Figure 5. Biofilm is dominated by long filamentous microbes and shows signs of mineralizing to become dental calculus (circle). Bar = 20 μm at an original magnification of 1,500x.

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Does caries initiate in the gingival sulcus?

- Gingival crevicular fluid: an exudate found in the gingival sulcus under inflammatory conditions
 - Composed of serum and locally generated materials (tissue breakdown products, inflammatory mediators, and antibodies directed against biofilm bacteria)
 - Flows from the depth of the sulcus up to the gingival margin
 - PH of 6.5-8.5
- Very unlikely for caries to grow in the gingival pocket as the pH of the GCF flushing out the sulcus is often over critical pH.
- Understand where caries is most likely to initiate as those are the sites you should pay extra attention to during your clinical exam.



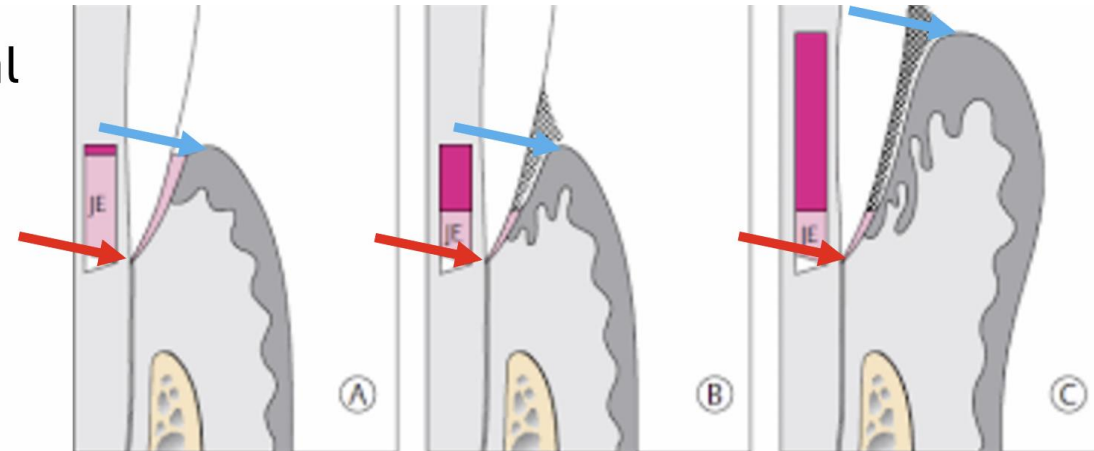
Pseudopockets

In cases where caries appears to have grown deep subgingivally, it is more likely to be due to the presence of pseudopockets.

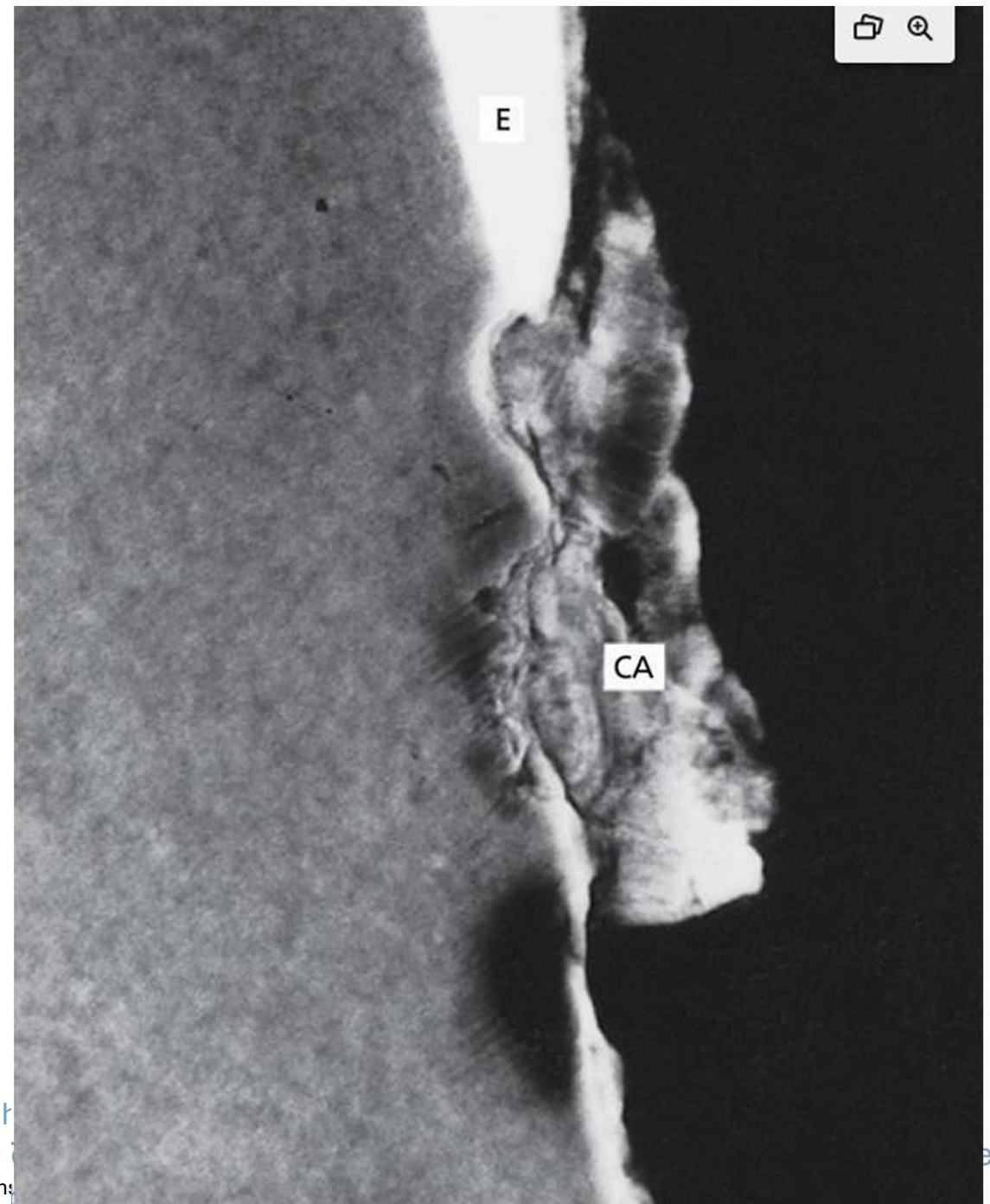
- Biofilm accumulates at the gingival margin undisrupted.
- In addition to caries, the biofilm may also cause gingival inflammation.
 - The coronal growth of the gingival margin due to inflammation forms a pseudopocket.
 - A "real" pocket is due to the apical advancement of the depth of the sulcus
- The swelling of the gingiva may give the impression that the lesion began in the periodontal pocket.

Blue = gingival margin

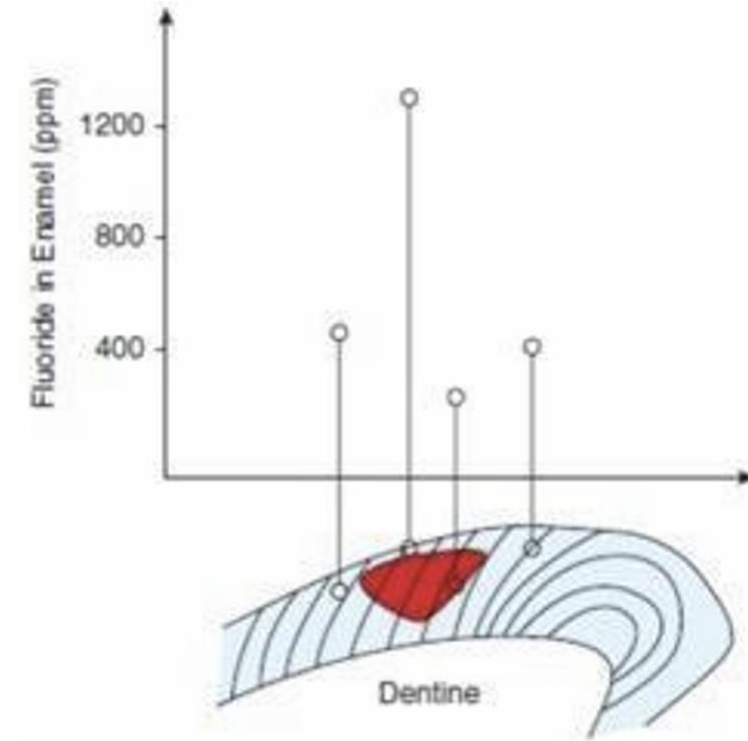
Red = depth of gingival sulcus



- Regular plaque removal from the surface of active root caries lesions is **not** likely to eliminate the micro organisms that have penetrated deep into the dentin.
- However, some lesions can be converted from **active into inactive** stages by non operative treatment.
- A change in the environmental conditions within the biofilm covering the root caries may result in mineral deposition.
 - o Calculus may be found partially occluding root surface defects corresponding to arrested carious lesions.
- What is calculus made of?
 - o Mineralized biofilm
 - Bacteria, HA, other minerals
- When does it form?
 - o In alkaline environments, which supported supersaturation in the saliva and promoted precipitation of minerals onto the tooth/biofilm.
 - o Do you think calculus causes caries?
- Where are the carious lesions in this image?



- Actively progressing root carious lesions can arrest with an increase in mineral content of the surface layer
 - Calculus
 - Fluorapatite
 - Remember the fluoride content is highest
 - In the remineralized structure over a carious lesion (whether enamel, dentin, or cementum)
 - Next highest in the remineralized structure adjacent to the lesion
 - Least F content deep within the lesion
- Regular localized application of topical fluoride is a common treatment modality for patients with root caries.
 - Often delivered through at home use of custom fluoride trays



Clinical Appearance

- Commonly found on interproximal root surfaces.
 - Look for protected sites, such as black triangles
- May not be black
 - "Softened" demineralized root structure often has slight darkening compared to healthy structure.
 - Darker and colder colors are sometimes reflective of arrested caries
- Dark calculus deposits can be mistaken for root caries.







Figure 2.26 Root surface caries lesions may start along the cervical region when gingiva starts to retract. No probing is needed.

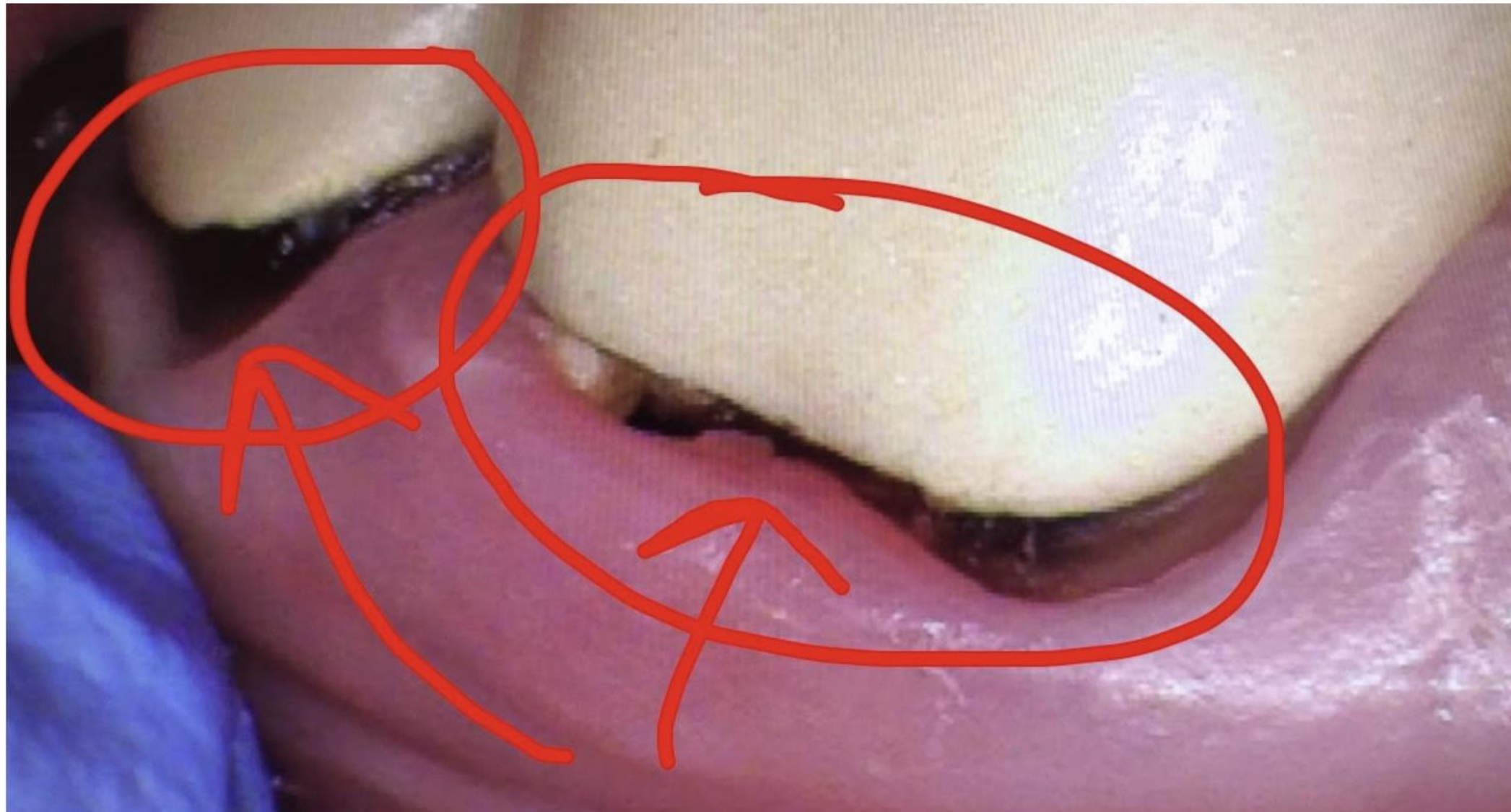


Figure 2.29 Inactive root surface caries lesions which are entirely dark and hard on probing. These are scar tissue resulting from previous active caries lesions! Imagine if a dentist had drilled and filled such lesions.

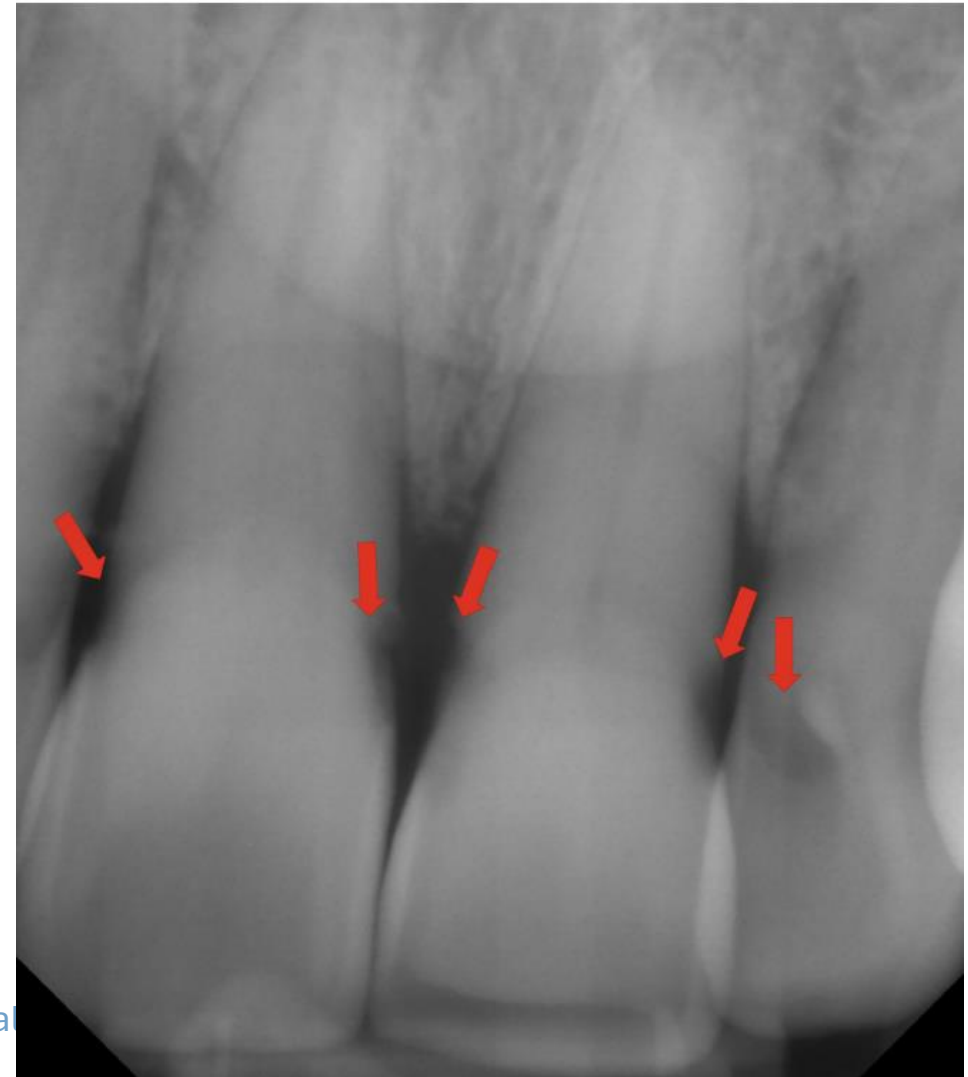


- Pg14 of textbook
- 3.50 and 3.51
 - White demineralized enamel
 - Leathery softened root structure
 - Do you see a lot of biofilm?
 - Do you see a lot of saliva?
 - Notice interproximal lesions too
- 3.52
 - Calculus
- 3.53
 - Softened root structure underneath calculus
 - However, high mineral content of calculus and dark appearance of lesions suggests some remineralization.
 - Arguably, these teeth may be maintainable with continued remin therapy.
 - What are some scenarios where that might be a reasonable treatment plan?





Radiographic Appearance



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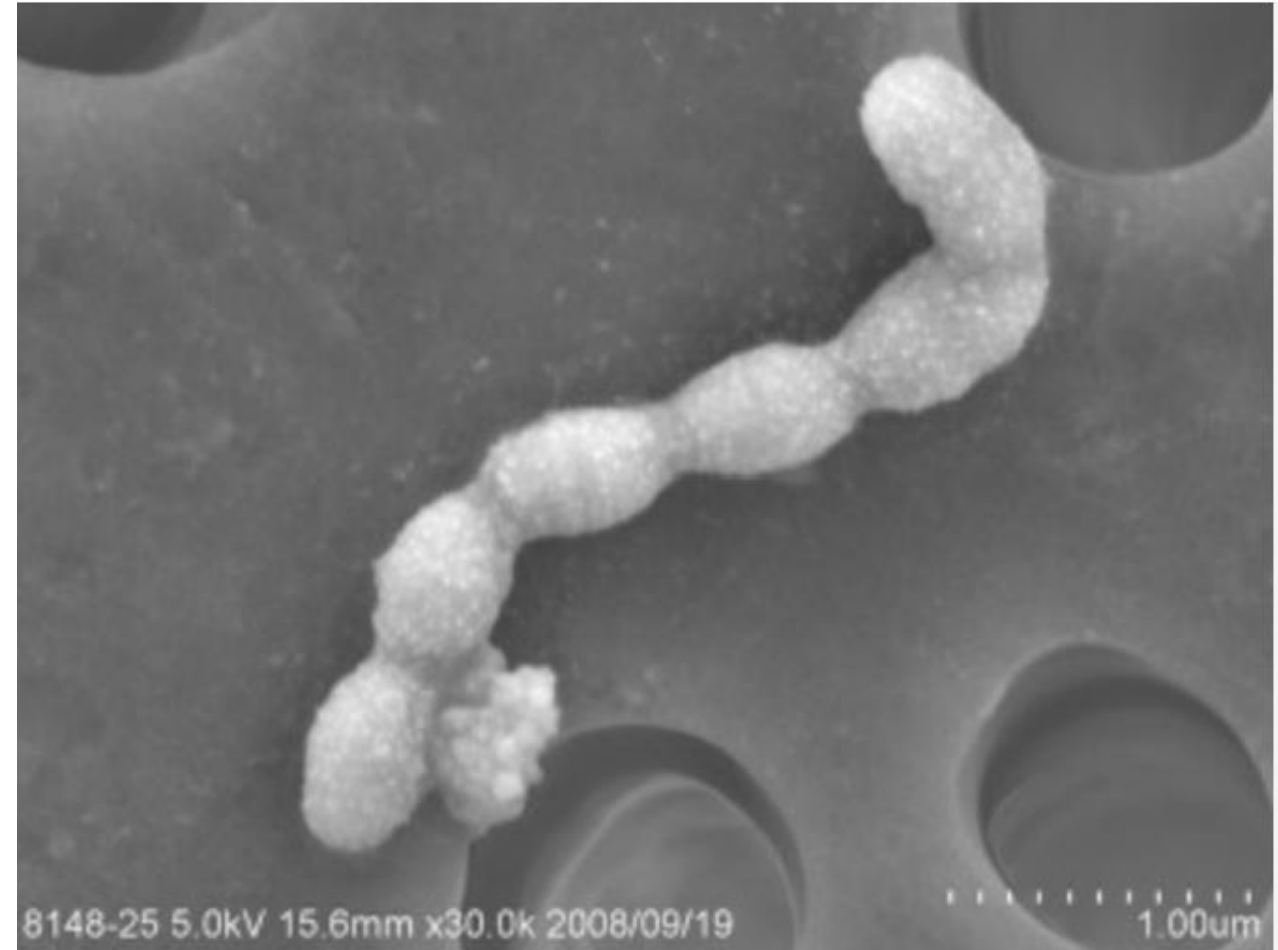
Remember to look
for "kissing lesions"
on adjacent tooth
surfaces.



Periapical x-ray showing root caries

Actinomyces

- Adheres well to exposed collagen of root surfaces (cementum and dentin) but not to enamel.
 - o Actinomyces is a late colonizer for enamel caries but can be an early colonizer for root caries.
- Prefers moderately acidic environments, matching the critical pH of cementum/dentin and thriving in the gingival sulcus environment
- Moderate acid production



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- Streptococcus mutans
 - Early colonizer and initiator of caries
 - Processes sucrose into glucans, producing EPS which:
 - Improves adhesion to enamel
 - Holds acid against enamel surface
- Lactobacillus
 - Associated with progression of caries more than initiation
 - Poor ability to adhere to intact enamel
 - Extremely acid tolerant and produces large amount of lactic acid
 - Thrives after cavitation occurs or in deep dentinal lesions
- On enamel
 - Streptococcus -> Actinomyces -> Lactobacillus
- On root
 - Actinomyces -> Streptococcus -> Lactobacillus

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Gingival Grafting

If your patient has gingival recession, particularly if they have other risk factors for root caries, consider whether gingival grafting may:

- Lower the patient's risk
- Be used in place of a direct restoration after excavation

Not all sites of recession are amenable to grafting. Consult with your periodontist.

As with all treatments, consider the etiology first.

Why did this patient have gingival recession in the first place?

If etiology is not addressed, the prognosis of ANY treatment (restoration, grafting, etc) will be compromised.

Figure 4: Teeth restored with Glass Ionomer filling.

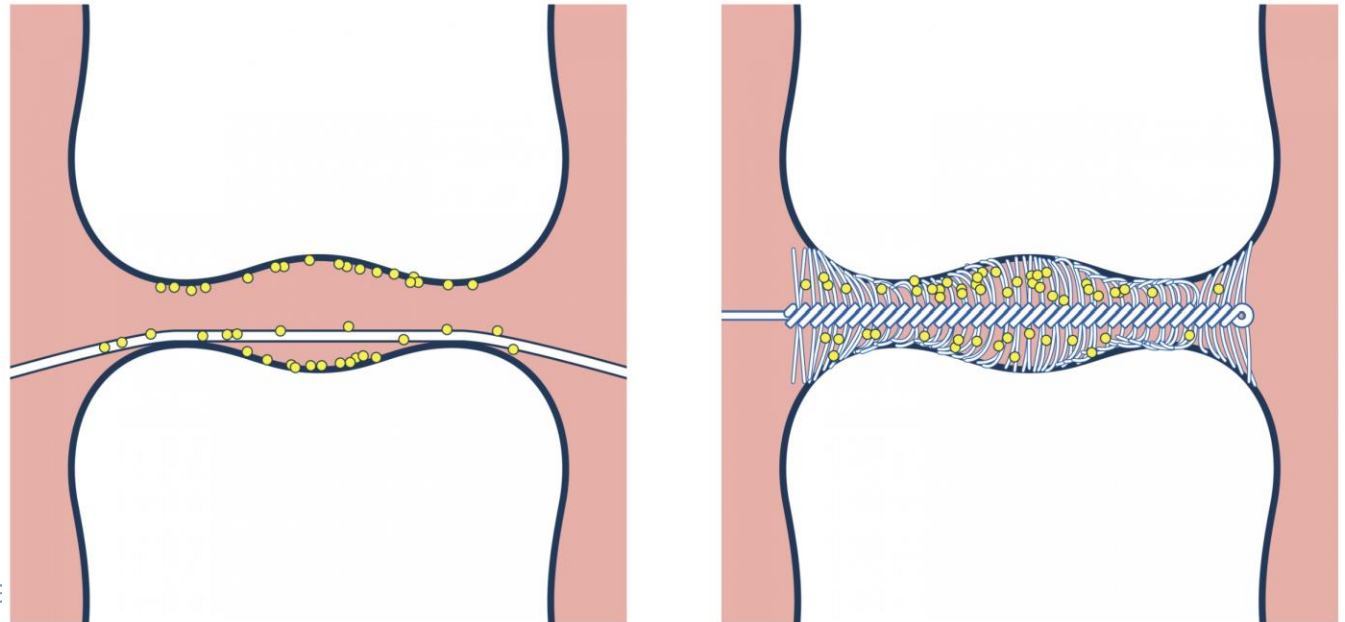


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Treating Root Caries

- Fluoride application to lower the critical pH of cementum/dentin
- Salivary stimulants preferred over salivary substitutes (discussed next week)
- Targeted hygiene instruments to better reach the protected sites associated with root exposure
 - Proxy brush
- Dental restorations



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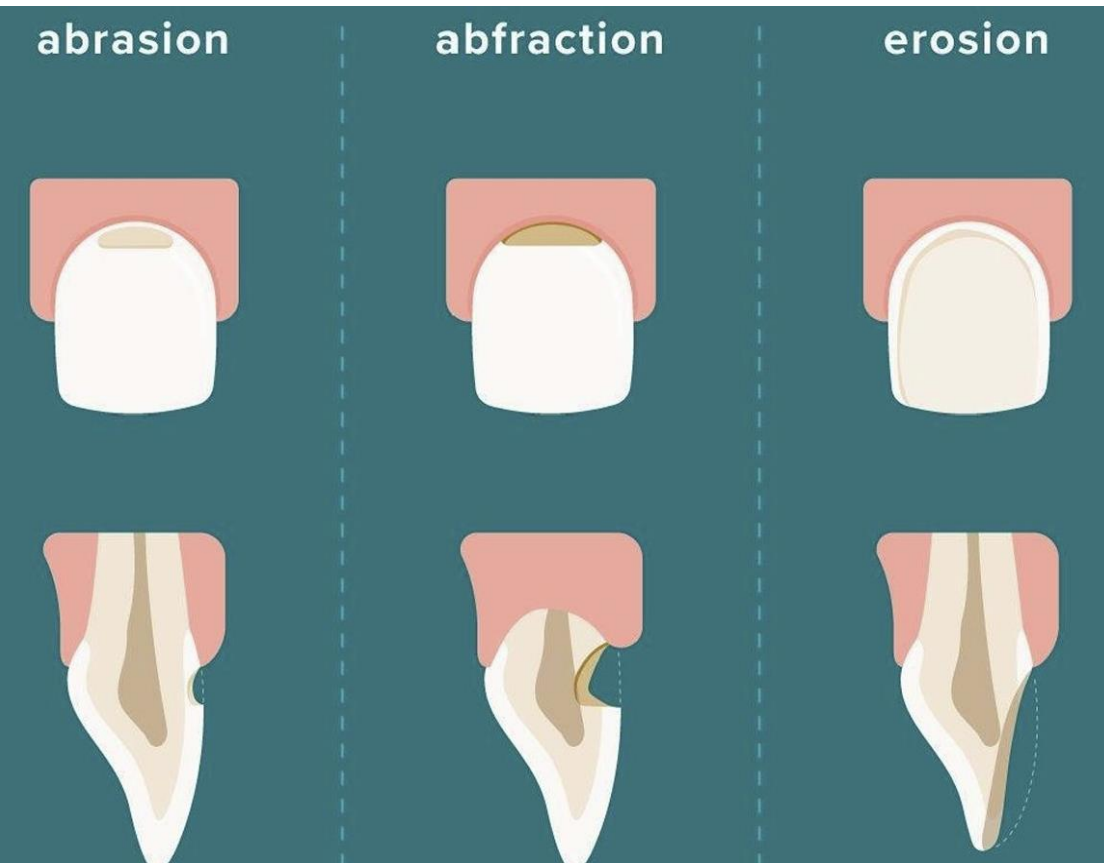
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Recurrent Root Caries

- Recurrent root caries are particularly prevalent because:
 - Etiologic factors are often difficult to address
 - Root caries may present even in cases with relatively little biofilm due to:
 - Xerostomia
 - Critical pH within lower range of physiologic pH
 - Bond strength to sclerotic dentin is significantly poorer than other types of restorations that typically bond to mostly enamel margins.
 - Micromechanical forces can promote loss or damage of cervical restorations, which often have minimal mechanical retention.
 - Is your restorative margin smoother or rougher than a CEJ?
- Inform your patient of the prognosis of each particular treatment and anything they can do to improve/worsen it. Even within the same patient, different restorations will have a different prognosis.



Common Cervical Root Pathologies

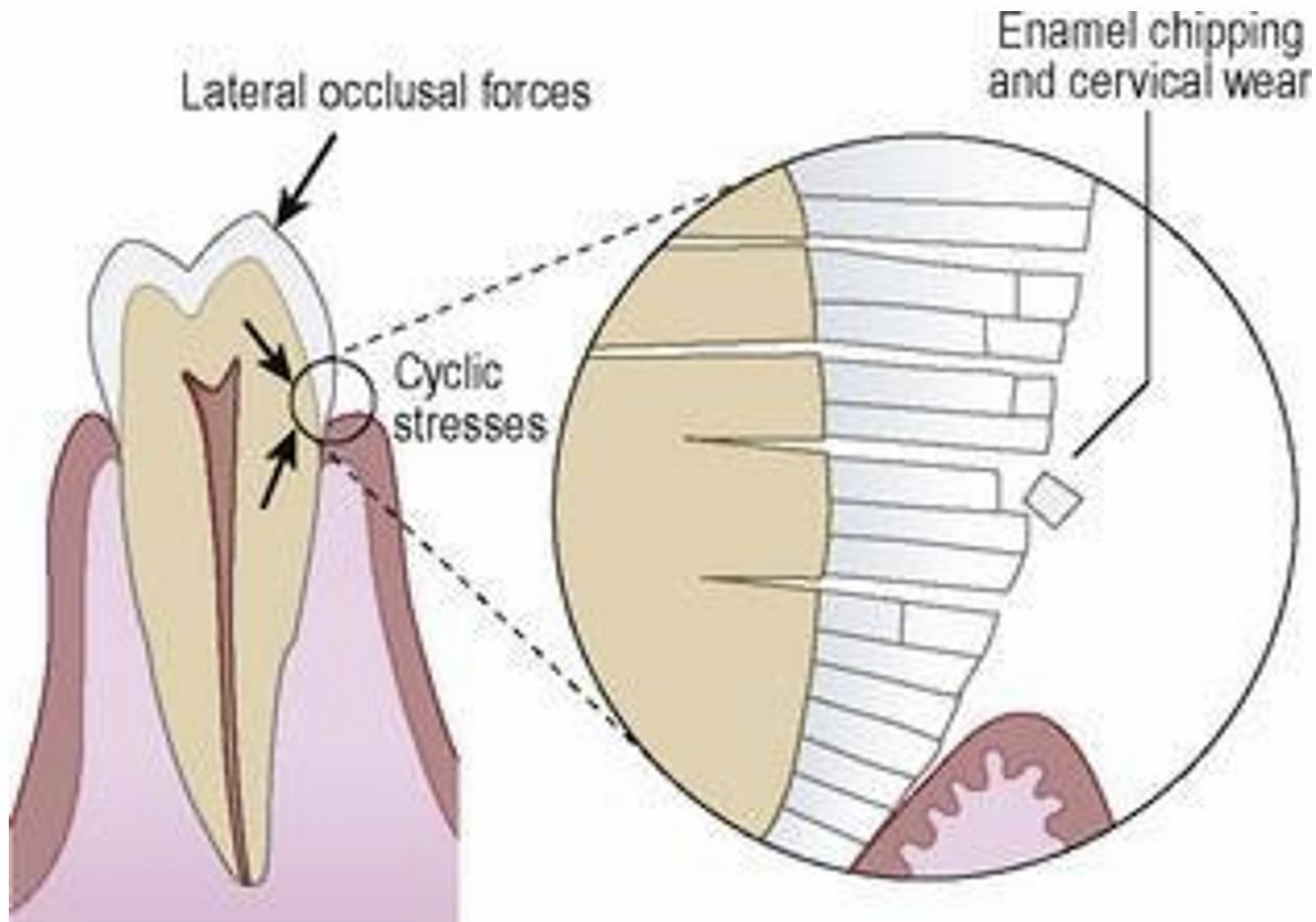


- Caries
- NCCLs (NOT caries)
 - Abrasion
 - Direct mechanical wear
 - Toothbrush, scaler, fingernail
 - Generally a rounded lesion.
 - May notice bevelling of the gingival margin.
 - Abfraction
 - Indirect mechanical wear through flexion
 - Generally a deeper triangular defect.
 - May extend beneath the gingival margin
 - Erosion
 - Direct chemical wear
 - Acidic drinks, reflux (possibly silent), sucking on lemons, excessive lozenge use.
 - Generally covers broad surface area, wears thin layers over time
 - Attrition: **Not** a NCCL.
 - Direct mechanical wear of opposing teeth (on occluding surfaces)
- External Cervical Resorption (NOT caries)
 - Covered in your endodontic course
 - Pathologic hyperactivity of osteoclasts that destroys tooth structure. Lesion is often red/pink from tissue infiltrate.

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NCCL- Abfraction



Abfraction

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NCCL - Erosion



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- Erosion is NOT caries because it is not biofilm mediated. It is acid mediated demineralization, but the acid does not come from the activity of a plaque biofilm.
- Enamel that is already demineralized from erosion:
 - Is thinner, weaker, and more likely to chip
 - We do NOT call this cavitation because it is not due to caries
 - Appears smooth but is microscopically porous
 - More plaque retentive
 - Less acid resistant because it is already demineralized
 - If caries does initiate, will cavitate more quickly since the enamel is already closer to the 30-40% threshold
- If all of the enamel is lost to erosion, dentin is exposed, which has a higher critical pH
- Erosion can therefore predispose a tooth to caries
- Notice:
 - Surface looks shiny
 - Loss of enamel is broad and thin
 - Dentin/DEJ shows through the thin enamel



NCCL - Abrasion



- Improper toothbrushing or scaling of root surfaces can damage or remove the cementum, exposing the dentin. Therefore, in real life, root surface caries often develop in exposed dentin.
- Abrasion is not caries
 - But abrasion can predispose a tooth to caries through loss of gingiva and root exposure
- Notice how thin the cementum is in the central part of the photo due to abrasion
 - Non abraded normal thickness cementum is seen at the most apical part of this image
- Why is the carious lesion positioned where it is? Where was this patient's gingival margin?

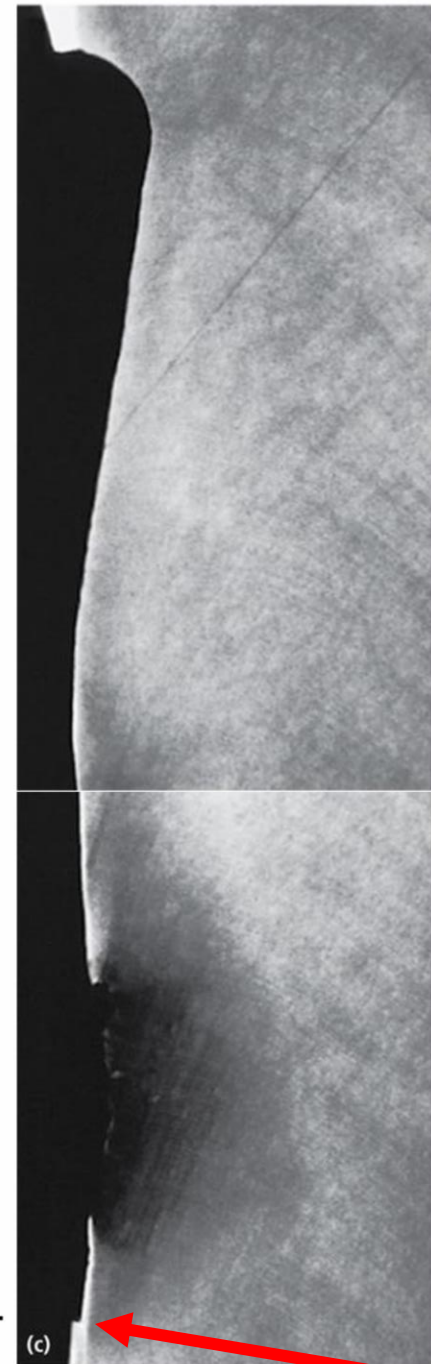
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CEJ

Apical

(c)



Mixed Lesions



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Resorption

Covered in your endodontic course in depth.

For now, notice that unlike caries, resorption:

- Doesn't look like cones of decay
- May initiate subgingivally
- May have striations or mothball eaten appearance
- May be present despite no biofilm and no acid.
- May have pink hue from soft tissue infiltrate
- May be associated with a history of trauma (though the pt may not be aware)
- **Should NOT be treated with a routine restoration like caries** – refer to Endodontist

